

Inequality and Business Cycles

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Motivation: Countercyclical inequality

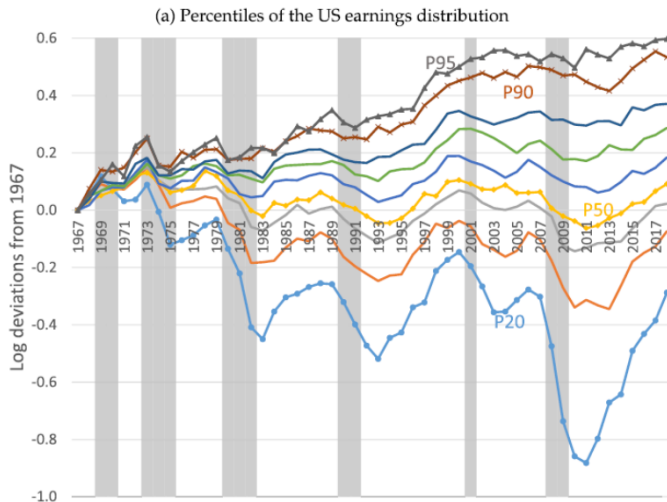
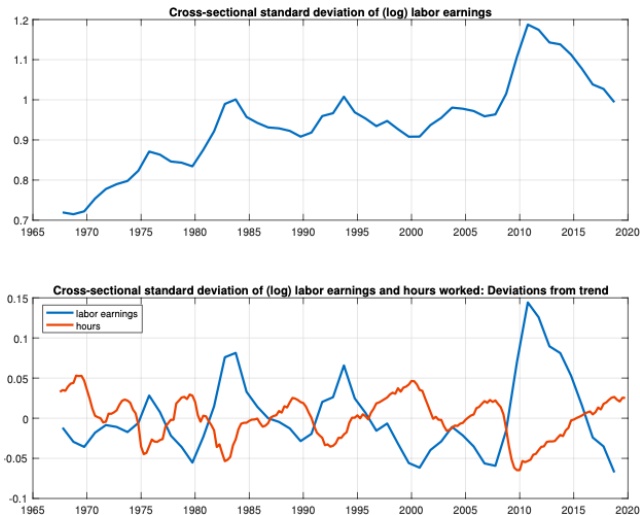


Figure 1: The evolution of the U.S. labor earnings distribution. Source: [Heathcote et al. \(2020\)](#).

Countercyclical hours and labour earnings



Source: Authors' computations, from FRED, Current Population survey and Bureau of Labor statistics

- Two questions:
 1. Does inequality affect business cycles?
 2. If so, which feature of inequality matters the most?
 - Countercyclicality?
 - Income risk?
- Method:
 - “Tractable” DSGE model with inequality and income risk
 - Use model to reproduce business cycles since 1950s
 - Shut down each feature of inequality and compare counterfactuals
- Key results:
 1. Inequality amplifies business cycles
 2. Main channel: Permanent inequality + countercyclical income risk
⇒ countercyclical precautionary saving motive ⇒ amplified cycles

Outline

- ① Model
- ② Estimation
- ③ Counterfactuals

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- Households heterogeneous in labour supplied and smoothing ability
- Intermediate producers hire both types of workers
- Calvo frictions: sticky wages and intermediate good prices
- Standard final good producer
- Fiscal policy and monetary authority
- Plethora of shocks and frictions, to best fit data

Household heterogeneity: S and H

- Households indexed by $h \in [0, 1]$
- Two types:
 - (1) Savers:
 - Measure $1 - \theta$
 - Smooth consumption over time
 - Labour complementary to capital
 - (2) Hand-to-mouth:
 - Measure θ
 - $MPC = 1$, always
 - Labour substitutable to capital

Household heterogeneity: Risk

- Probability to be S tomorrow if you are S today:

$$Pr(h_{t+1} = S | h_t = S) = s_t$$

- $Pr(h_{t+1} = H | h_t)$ set so that measures remain constant
- Switching probability depends on GDP gap:

$$s_t = s_0 \left(\frac{X_t}{X_t^*} \right)^{s_1}$$

s_0 : Steady-state probability

s_1 : Elasticity of s_t wrt GDP gap

X_t : $GDP = C_t + (1 - \theta)I_{S,t} + G_t$

X_t^* : Potential GDP (Curdia et al., 2015)

Instantaneous utility

- Households supply differentiated labour types $j \in [0, 1]$
- Labour supply set by a h union, given Calvo friction Wage setting
- For $h \in \{S, H\}$:

$$u_{h,t} = b_t \left[\log (C_{h,t} - \eta_h C_{h,t-1}) + \varphi_h \int_0^1 \frac{L_{h,t}(j)^{1+\nu_h}}{1+\nu_h} dj \right]$$

b_t : Intertemporal preference shock

η_h : External habit formation parameter

φ_h : Consumption/leisure relative preference

ν_h : Inverse Frisch elasticity of labour

Savers smooth consumption

- Objective function:

$$u_{S,t} + \beta \mathbb{E}_t [s_{t+1} u_{S,t+1} + (1 - s_{t+1}) u_{H,t+1}]$$

- Budget constraint:

$$P_t(C_{S,t} + I_{S,t}) + B_{S,t} \leq \int_0^1 W_{S,t}(j) L_{S,t}(j) dj + \bar{K}_{S,t-1} [r_t^k u_t - P_t a(u_t)] \\ + s_{t-1} R_{t-1} s_t B_{S,t-1} + \Pi_{S,t} + T_{S,t}$$

- Physical capital accumulation equation:

$$\bar{K}_{S,t} = (1 - \delta) \bar{K}_{S,t-1} + \mu_t \left[1 - \psi \left(\frac{I_{S,t}}{I_{S,t-1}} \right) \right] I_{S,t}$$

- Effective vs physical capital:

$$K_{S,t} = u_t \bar{K}_{S,t-1}$$

Savers' Euler equation

- FOC[$B_{S,t}$]:

$$\Lambda_{S,t} = \beta \varsigma_t R_t \mathbb{E}_t \{ s_{t+1} \Lambda_{S,t+1} + (1 - s_{t+1}) \Lambda_{H,t+1} \}$$

- $\Lambda_{h,t}$: Marginal utility of income of type h
- If $\Lambda_H > \Lambda_S$ and $s_{t+1} < 1$: precautionary saving motive
- If s_{t+1} is countercyclical: countercyclical saving motive
- If Λ_H countercyclical: amplified precautionary motive

Marginal Utilities of Income

Hand-to-mouths eat it all

- $MPC = 1$, hence:

$$P_t C_{H,t} = \int_0^1 W_{H,t}(j) L_{H,t}(j) + \frac{(1-\theta)(1-s_t)B_{S,t-1}}{\theta} s_{t-1} R_{t-1} + T_{H,t}$$

- Bond holdings of time $t-1$ savers who became hand-to-mouth at time t equally distributed among time t hand-to-mouths

- Intermediate good producers
 - KORV (Krusell et al., 2000) production function:

$$Y_t(i) = \left\{ \omega [A_t \mathcal{L}_{H,t}(i)]^\sigma + (1 - \omega) \left[\alpha a_{k,t} K_t^\varrho(i) + (1 - \alpha) [A_t \mathcal{L}_{S,t}(i)]^\varrho \right]^{\frac{\sigma}{\varrho}} \right\}^{\frac{1}{\sigma}} - A_t F$$

- $\mathcal{L}_H$ and K_t : substitutes
 - $\mathcal{L}_S$ and K_t : complements
- Final good producer is standard, with markup shocks. Final good producer

- Fiscal policy: set G_t , $T_{H,t}$ and $T_{S,t}$
 - Balanced budget: $\theta T_{H,t} + (1 - \theta) T_{S,t} + P_t G_t = 0$
 - Bonds are in zero net supply
 - Transfers and taxes implicitly defined:

$$\frac{\mathcal{I}_{H,t} + T_{H,t}}{\mathcal{I}_{S,t} + T_{S,t}} = \tau_0 \left(\frac{\mathcal{I}_{H,t}/\mathcal{I}_{S,t}}{\mathcal{I}_H/\mathcal{I}_S} \right)^{\tau_1}$$

$\mathcal{I}_{h,t}$: Pre-tax income of representative household of type h

τ_1 : Volatility in income inequality (no volatility if 0)

τ_0 : Level of income inequality

- Monetary policy:

$$\frac{R_t}{R} = \left(\frac{R_{t-1}}{R} \right)^{\rho_R} \left[\left(\frac{\prod_{j=0}^3 \pi_{t-j}}{\pi} \right)^{\phi_\pi} \left(\frac{X_t}{X_t^*} \right)^{\phi_X} \right]^{1-\rho_R} \left(\frac{X_t/X_t^*}{X_{t-1}/X_{t-1}^*} \right)^{\phi_{dX}} \eta_{mp,t}$$

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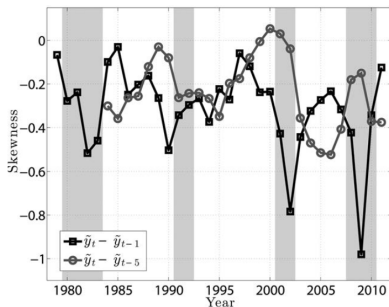
Key parameters

- Data from FRED, CPS and Bureau of Labor Statistics

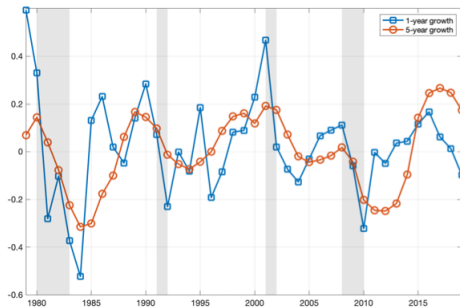
Parameter	Value	Remark
<i>Estimated</i>		
σ	9.38	$\mathcal{L}_H$ and K substitutes
ϱ	0.38	$\mathcal{L}_S$ and K complements
s_1	0.368	Countercyclical risk
τ_0	0.35	Permanent inequality
τ_1	0.21	Cyclical inequality
<i>Calibrated</i>		
θ	0.2	Parker et al. (2013), etc.
s_0	0.987	Guvenen et al. (2021)

Income risk: Data vs estimated model

- Skewness of 1- and 5- year changes in log annual labour earnings
- Large drops in income more likely in recessions
- Left skewness of distribution is countercyclical

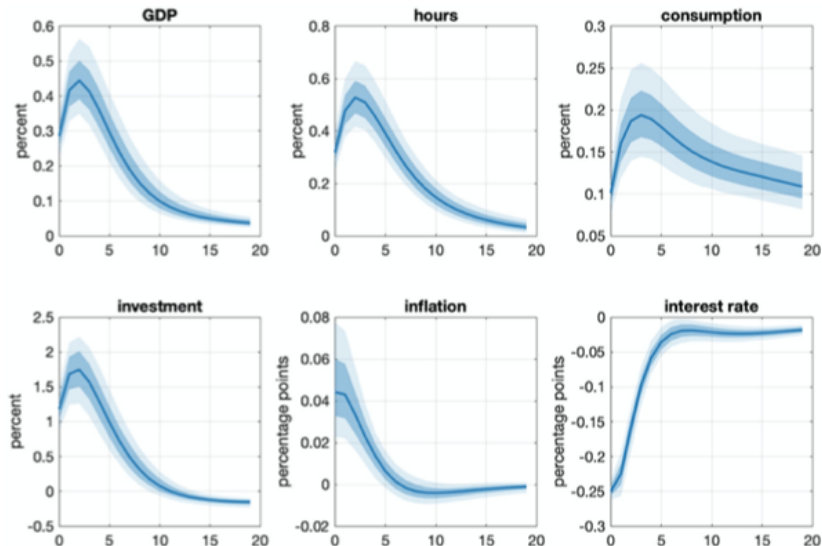


(a) Data (Guvenen et al., 2014)

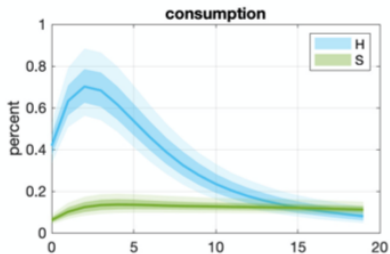
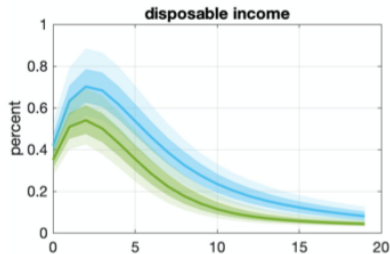
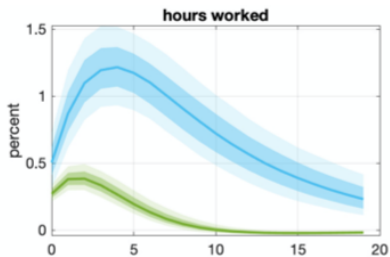
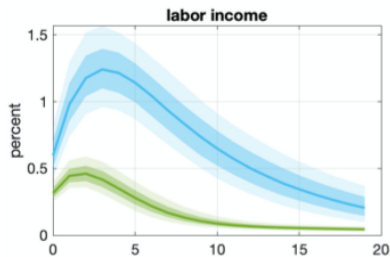


(b) Estimated model

Aggregate IRFs to monetary shock



Type-specific IRFs to monetary shock



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Counterfactual experiments: Method

- Use estimated model to get disturbances behind business cycles
- Take these disturbances as fixed
- Shut down features of the model to assess their significance in explaining business cycles

Four ways inequality and business cycles interact

- Cyclical inequality:

$$\Lambda_{H,t} \neq \Lambda_{S,t}$$

- Permanent inequality:

$$\Lambda_H \neq \Lambda_S$$

- Cyclical idiosyncratic risk:

$$s_1 > 0$$

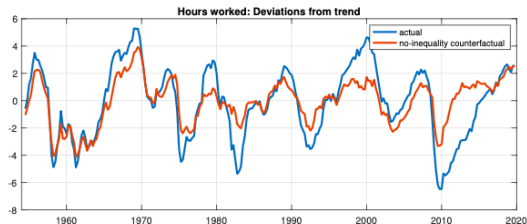
- Permanent idiosyncratic risk:

$$s_0 < 1$$

→ Goal: shut down each channel one at a time

Experiment 0: Overall role of inequality

- Impose $\Lambda_{H,t} = \Lambda_{S,t}$ and $\Lambda_H = \Lambda_S$ as equilibrium condition
- Inequality matters for business cycles



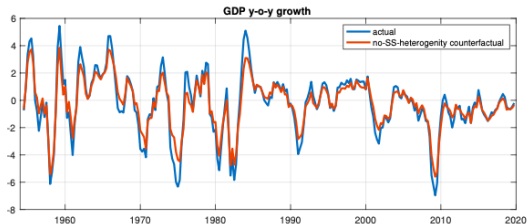
Experiment 1: No cyclical inequality

- Impose $|\Lambda_{S,t} = \Lambda_{H,t}| = |\Lambda_S - \Lambda_H|$ with transfers
- Cyclical inequality not relevant for business cycles



Experiment 2: No permanent inequality

- Impose $\Lambda_S = \Lambda_H$, with transfers
- Significant fall in volatility



Experiment 3: No cyclical idiosyncratic risk

- Set $s_t = s_0$ always
- Fall in volatility



Experiment 4: No permanent idiosyncratic risk

- Set $s_t = 1$ always
- No change from above



Overview of counterfactual analysis

- First line: Standard deviation of GDP growth
- Second line: Standard deviation of detrended hours
- In bold: Baseline

Risk		Inequality			
		Only SS	Only cyclical	Both	Neither
		$\Lambda_{H,t} = \Lambda_{S,t}$ $\Lambda_H \neq \Lambda_S$	$\Lambda_{H,t} \neq \Lambda_{S,t}$ $\Lambda_H = \Lambda_S$	$\Lambda_{H,t} \neq \Lambda_{S,t}$ $\Lambda_H \neq \Lambda_S$	$\Lambda_{H,t} = \Lambda_{S,t}$ $\Lambda_H = \Lambda_S$
Only SS	$s_0 = 0.987$	1.65			
	$s_1 = 0$	2.82			
Both	$s_0 = 0.987$	2.19	1.75	2.25	1.64
	$s_1 = 0.367$	3.3	1.65	2.61	1.66
Neither	$s_0 = 1$	1.65			
	$s_1 = 0$	2.82			

- How do business cycles and inequality interact?
- Inequality amplifies business cycles
- Two channels:
 - (1) Permanent inequality
 - (2) Countercyclical income risk
- Together, make precautionary savings countercyclical, thereby amplifying business cycles

A1: Marginal Utility of Income

- FOC[$B_{S,t}$]:

$$\Lambda_{S,t} = \beta \varsigma_t R_t \mathbb{E}_t \{ s_{t+1} \Lambda_{S,t+1} + (1 - s_{t+1}) \Lambda_{H,t+1} \}$$

- Where:

$$\Lambda_{S,t} = \frac{b_t}{P_t(C_{S,t} - \eta_S C_{S,t-1})}$$

$$\Lambda_{S,t+1} = \frac{s_{t+1} b_{t+1}}{P_{t+1}(C_{S,t+1} - \eta_S C_{S,t})}$$

$$\Lambda_{H,t+1} = \frac{1 - \theta}{\theta} \frac{b_{t+1}(1 - s_{t+1})}{P_{t+1}(C_{H,t+1} - \eta_H C_{S,t})}$$

A2: Union h 's problem

- Given firm's labour demand $L_{h,t}$, demand for labour j solves:

$$\begin{aligned} \max_{\{L_{h,t}(j)\}_{j=0}^1} \quad & W_{h,t} L_{h,t} - \int_0^1 W_{h,t}(j) L_{h,t}(j) dj \\ \text{s.t.} \quad & L_{h,t} = \left(\int_0^1 L_{h,t}(j)^{\frac{1}{1+\lambda_{w,t}}} dj \right)^{1+\lambda_{w,t}} \end{aligned}$$

- Demand for labour of type j is standard:

$$L_{h,t}(j) \equiv \left(\frac{W_{h,t}(j)}{W_{h,t}} \right)^{-\frac{1+\lambda_{w,t}}{\lambda_{w,t}}} L_{h,t}$$

- Aggregate wage index:

$$W_{h,t} = \left(\int_0^1 W_{h,t}(j)^{-\frac{1}{\lambda_{w,t}}} dj \right)^{-\lambda_{w,t}}$$

A3: h -specific unions set wages given Calvo pricing

$$\begin{aligned} \max_{W_{h,t}(j)} \quad & \mathbb{E}_t \left\{ \sum_{s=0}^{\infty} (\xi_w \beta)^s \left[\Lambda_{h,t+s} W_{h,t}(j) \Pi_{t,t+s}^w L_{h,t,t+s}(j) - u_{h,t}(j) \right] \right\} \\ \text{s.t.} \quad & u_{h,t}(j) = b_t \left[\log(C_{h,t} - \eta_h C_{h,t-1}) + \varphi_h \frac{L_{h,t}(j)^{1+\nu_h}}{1+\nu_h} \right] \\ & L_{h,t,t+s}(j) = \left(\frac{W_{h,t}(j) \Pi_{t,t+s}^w}{W_{h,t+s}} \right)^{-\frac{1+\lambda_{w,t+s}}{\lambda_{w,t+s}}} L_{h,t+s} \\ & \Pi_{t,t+s}^w = \prod_{k=1}^s \left[(\pi e^\gamma)^{1-\iota_w} (\pi_{t+k-1} e^{z_{t+k-1}})^{\iota_w} \right] \end{aligned}$$

- Utility of worker j is weighted
- Yields two wage Phillips curves: one for $L_{H,t}$, and one for $L_{S,t}$

A4: Final good producer

- Demand for $Y_t(i)$:

$$Y_t(i) = \left(\frac{P_t(i)}{P_t} \right)^{-\frac{1+\lambda_{p,t}}{\lambda_{p,t}}} Y_t$$

- Price of Y_t :

$$P_t = \left[\int_0^1 P_t(i)^{-\frac{1}{\lambda_{p,t}}} di \right]^{-\lambda_{p,t}}$$

- Final good:

$$Y_t = \left[\int_0^1 Y_t(i)^{\frac{1}{1+\lambda_{p,t}}} di \right]^{1+\lambda_{p,t}}$$

A5: Wage setting

- Demand for labour of type j : Labour demand

$$L_{h,t}(j) \equiv \left(\frac{W_{h,t}(j)}{W_{h,t}} \right)^{-\frac{1+\lambda_{w,t}}{\lambda_{w,t}}} L_{h,t}$$

- Aggregate wage index:

$$W_{h,t} = \left(\int_0^1 W_{h,t}(j)^{-\frac{1}{\lambda_{w,t}}} dj \right)^{-\lambda_{w,t}}$$

- Representative union h sets wage given Calvo friction Union problem

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